Cryptocurrency Price Direction Prediction Using Market Indicators

Project Report — Data Collection, Integration and Preprocessing

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Course	Data Collection, Integration and Preprocessing (CIP02)
Semester	FS 2026
Group	103
Date	April 2026

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Abstract. This report presents a binary classification pipeline to predict the daily price movement direction (up or down) of three major cryptocurrencies (Bitcoin, XRP, and Internet Computer) using publicly available market indicators from the preceding day. Historical OHLCV data spanning three years was collected from CoinMarketCap using a Playwright-based scraper, cleaned, and enriched with log-transformed technical features. An exploratory data analysis confirmed near-random return dynamics and identified a five-feature stationary predictor set. Two classifiers were evaluated under a strict chronological 70/15/15 train/validation/test split: Logistic Regression and Decision Tree. Test-set accuracy ranged from 40 to 63 %, consistent with the efficient market hypothesis and the near-random walk behaviour identified in the EDA.

1. Introduction

Cryptocurrency markets operate continuously and are characterised by high volatility, rapid price swings, and strong sensitivity to market sentiment. These properties make them an attractive but challenging domain for short-term predictive modelling. Understanding whether tomorrow's price direction can be anticipated from today's observable market indicators has direct relevance for trading strategies and risk management.

This project investigates whether publicly available daily market data for three major cryptocurrencies, Bitcoin (BTC), Ripple (XRP), and Internet Computer (ICP), can be used to predict the direction of next-day price movement. The full pipeline covers data collection, preprocessing, exploratory data analysis, and binary classification using machine learning.

Three research questions guide the work:

1. Which market indicators have the strongest relationship with daily cryptocurrency price changes?
2. How accurately can historical indicators from day $t-1$ predict price direction on day t?
3. Which machine learning model achieves the highest classification accuracy?

2. Methods

2.1. Data Source and Collection

Historical OHLCV data for Bitcoin, XRP, and ICP were collected from **CoinMarketCap** using a Playwright-based scraper (`src/coinmarket_scrape.py`). Playwright drives a headless Chromium browser to handle the JavaScript-rendered price tables. The scrape covers daily records from 2022-12-08 to 2026-03-19, yielding **1 198 rows per coin** The seven fields are date, open, high, low, close, 24-hour volume, and market capitalisation.

2.2. Preprocessing and Feature Engineering

Raw files were inspected for missing values, duplicate dates, and OHLCV constraint violations. All checks passed without issues. No outliers were removed. Extreme price events such as exchange listings or market crashes are real market signals that must be preserved.

Ten log-transformed features were computed per coin. **Logarithmic scaling** was applied throughout because cryptocurrency prices are log-normally distributed, log returns are symmetric and additive, and log-transforming volume and market capitalisation removes scale dominance across the three assets.

A **stationarity analysis** showed that level features (`log_close`, `ma_7`, `ma_30`, `log_market_cap`) are non-stationary and exhibit pairwise Pearson correlations of $r \geq 0.91$. Including them as predictors would produce spurious correlations with the target. They were therefore excluded, leaving five stationary features for modelling (Table 1).

Table 1: Feature set used for modelling (all values from day $t-1$).

Feature	Description
`log_return`	$\log(p_t/p_{t-1})$ — daily log return
`log_close_open_ratio`	log(close/open) — intraday direction
`log_high_low_ratio`	log(high/low) — intraday range
`volatility_7`	7-day rolling standard deviation of log returns
`log_volume_change`	$\log(v_t/v_{t-1})$ — relative volume change

All features are lagged by one day before training so the model only observes information available prior to the prediction date. The binary target `price_direction` is 1 if $\log(p_t/p_{t-1}) > 0$, else 0.

2.3. Analysis Methods

Exploratory analysis covered autocorrelation (ACF for lags 1 to 20), lag-1 Pearson correlations between each feature and the target, cross-asset log return correlations, and rolling volatility.

Modelling uses two classifiers evaluated under a strict chronological **70 / 15 / 15 train/validation/test split** applied independently per coin. Shuffling is not performed, as it would allow future data to enter the training set. The models are Logistic Regression (`LogisticRegression(max_iter=1000)`) and Decision Tree (`DecisionTreeClassifier()`). Model complexity was kept deliberately low to match the weak signal structure identified in the EDA. Each model is evaluated on accuracy, precision, recall, and F1-score, with accuracy as the primary metric given the near-balanced class distribution.

3. Discussion

3.1. RQ1 — Which Indicators Relate Most Strongly to Price Changes?

To answer the first research question, lag-1 Pearson correlations were computed between each stationary feature at day $t-1$ and `price_direction` at day t. Non-stationary level features (`log_close`, `ma_7`, `ma_30`, `log_market_cap`) were excluded beforehand. They trend with the price level and exhibit pairwise correlations of $r \geq 0.91$, which would produce spurious associations with the target.

The strongest statistically significant predictor for Bitcoin and XRP is `log_return` ($r = -0.066$, $p = 0.023$ and $r = -0.081$, $p = 0.006$ respectively), indicating a weak mean-reversion tendency. A positive return yesterday is slightly more likely to be followed by a downward move today. For ICP the significant predictor is `volatility_7` ($r = -0.062$, $p = 0.034$), suggesting that elevated recent volatility weakly predicts a downward direction. All r^2 values remain below 0.7 %, confirming that the relationships are statistically detectable but practically negligible. Figure 1 visualises the full lag-1 correlation profile for each coin.

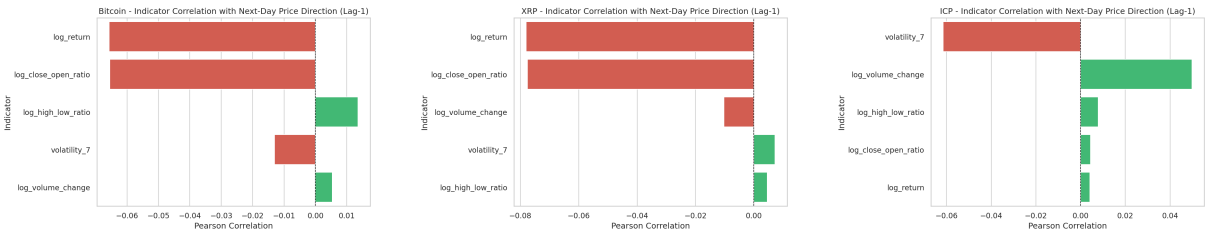

Figure 1: Lag-1 Pearson correlation between each feature at $t-1$ and `price_direction` at t for Bitcoin (left), XRP (centre), and ICP (right). Bars marked with * ($p < 0.05$) or ** ($p < 0.01$) are statistically significant. All magnitudes are small, confirming weak predictive power across all features and coins.

3.2. RQ2 — How Accurately Can Indicators Predict Price Direction?

ACF analysis sets the predictability ceiling. The autocorrelation function (ACF) measures whether past returns linearly predict future returns. Under the weak-form Efficient Market Hypothesis, all ACF coefficients would be zero. Figure 2 shows the ACF for lags 1 to 20 for all three coins, with 95 % confidence bands of $\pm 1.96/\sqrt{n} = \pm 0.057$.

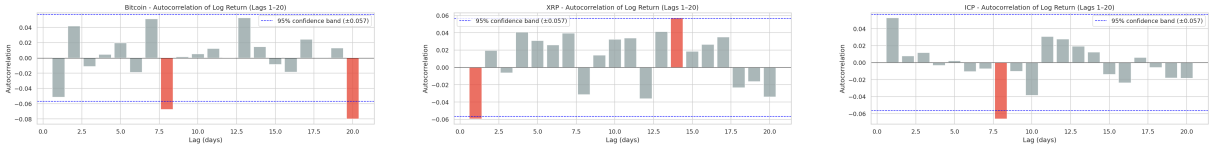

Figure 2: ACF of daily log returns for Bitcoin (left), XRP (centre), and ICP (right), lags 1 to 20. Dashed lines mark the 95 % confidence band (±0.057). Nearly all coefficients fall within the band, confirming near-random walk behaviour. XRP shows a slightly negative lag-1 coefficient, consistent with the mean-reversion signal in Figure 1.

Nearly all coefficients fall within the confidence band across all three coins, confirming near-random walk behaviour. The accuracy ceiling of 52 to 58 % is derived from three observations. First, the class balance sets the trivial floor. A classifier that always predicts the majority class achieves 50.4 % (BTC), 51.1 % (XRP), and 52.6 % (ICP) without learning anything. Second, the ACF confirms the series is statistically consistent with white noise, so the realistic accuracy is very close to this floor. Third, the significant lag-1 correlations (max $|r| = 0.081$ for XRP) provide a rough upper bound. For a binary target with a single predictor of correlation r and class probability $P \approx 0.5$, the maximum accuracy gain from a linear classifier is approximately $|r| \cdot \sqrt{P(1-P)} \approx 0.081 \times 0.5 \approx 4$ percentage points above the baseline, placing the theoretical ceiling at roughly 54 to 58 %. Any result substantially above this range would indicate data leakage, which was confirmed when initial experiments with unlagged features produced 100 % Decision Tree accuracy.

Rolling volatility and market regimes. Figure 3 shows the 30-day rolling standard deviation of log returns for all three coins. Volatility is not constant over time. Distinct high-volatility regimes are visible, particularly in 2022 and early 2024. ICP consistently shows the highest volatility (mean $\sigma_7 = 4.41$ %), followed by XRP (3.41 %) and Bitcoin (2.23 %). This time-varying structure suggests that a static model trained on the full period may not capture regime-specific dynamics, which is a further reason why accuracy is limited.

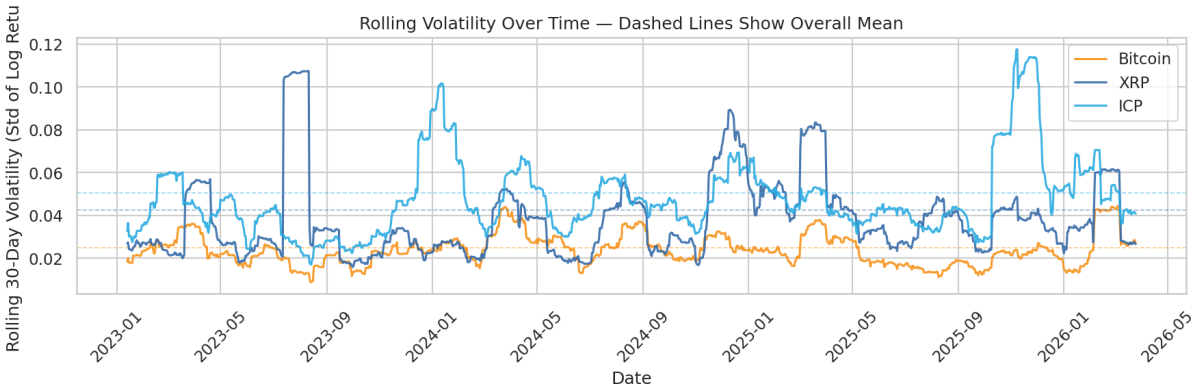

Figure 3: 30-day rolling standard deviation of log returns for all three coins. Volatility clustering and distinct market regimes are visible. ICP consistently exhibits the highest volatility throughout the period.

Cross-asset co-movement. Figure 4 shows moderate log return correlations across coins (BTC-XRP $r = 0.589$, BTC-ICP $r = 0.572$, XRP-ICP $r = 0.491$). Separate models were trained per coin to preserve coin-specific dynamics.

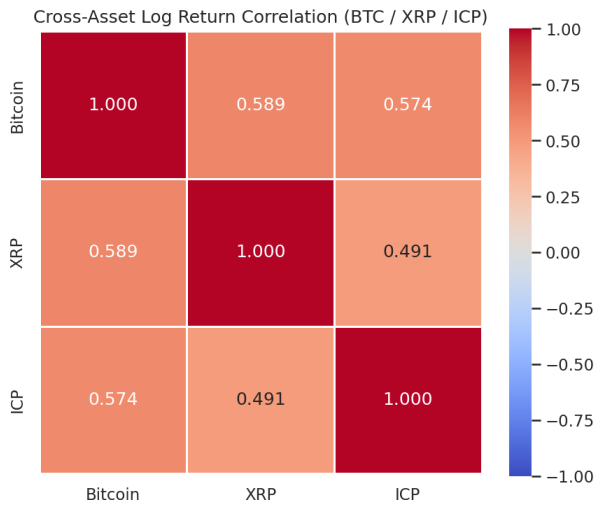

Figure 4: Cross-asset log return correlation heatmap (Pearson r). Co-movement is present but substantial coin-specific variance justifies separate models per coin.

3.3. RQ2 & RQ3 — Model Evaluation and Model Comparison

Table 2 and Figure 5 report test-set performance for both models across all three coins.

Table 2: Test-set evaluation (70/15/15 chronological split). Random baseline = 0.50. Realistic ceiling from ACF = 0.58.

Coin	Model	Acc.	Prec.	Rec.	F1
Bitcoin	Logistic Reg.	0.453	0.455	0.698	0.551
	Decision Tree	0.447	0.427	0.442	0.434
ICP	Logistic Reg.	0.626	0.625	0.141	0.230
	Decision Tree	0.503	0.390	0.451	0.418
XRP	Logistic Reg.	0.397	0.372	0.786	0.505
	Decision Tree	0.436	0.358	0.557	0.436

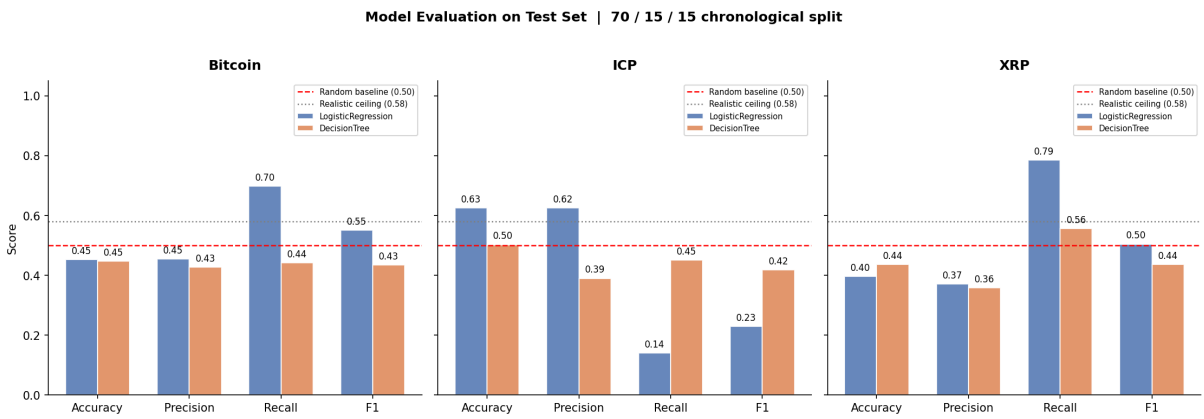

Figure 5: Test-set scores per coin and model. The red dashed line marks the 0.50 random baseline. The grey dotted line marks the 0.58 ACF-derived ceiling.

RQ2. All accuracy values lie between 40 and 63 %, consistent with the ceiling established by the ACF analysis. This confirms that yesterday's market indicators are insufficient for reliable next-day price direction prediction.

RQ3. Neither model consistently outperforms the other across all three coins. Logistic Regression is marginally more stable. The ICP accuracy of 62.6 % for Logistic Regression appears high but is undermined by a recall of only 14.1 %, revealing that the model almost exclusively predicts the majority class rather than learning a genuine signal. The Decision Tree gains no benefit from its higher capacity, confirming that the data lacks the non-linear structure required to justify a more complex model.

4. Conclusion

Summary and Learnings

This project built a reproducible end-to-end pipeline from raw web scraping to binary classification for cryptocurrency price-direction prediction. The central finding is that publicly available daily market indicators have very limited predictive power for next-day price direction. Test-set accuracy for both models and all three coins is consistent with random-walk behaviour and aligns with the theoretical ceiling derived from the ACF analysis.

Three methodological lessons stood out. First, stationarity matters. Including non-stationary level features such as moving averages and market capitalisation introduced spurious correlations that would have inflated reported performance. Second, data leakage is easy to introduce in time-series settings. Initial experiments with unlagged features produced 100 % Decision Tree accuracy, which immediately revealed the error. Third, the EDA correctly predicted the performance range before any model was trained, demonstrating that thorough exploratory analysis is a prerequisite for trustworthy modelling.

Limitations

- The feature set is limited to price-derived technical indicators. Sentiment data, on-chain metrics, and macroeconomic variables are excluded.
- Only two classifiers are evaluated. Ensemble methods such as Random Forest may offer marginal improvements without additional data.
- The Decision Tree is not hyperparameter-tuned. Pruning could reduce the gap between validation and test performance.

Outlook

Future work could enrich the feature set with news sentiment or social media signals, apply TimeSeriesSplit cross-validation for more robust model selection, or investigate whether predictability increases over multi-day prediction horizons. Extending the pipeline to additional coins would allow broader generalisations about market efficiency across different asset classes.

Tools & Libraries: Python 3.10+, Playwright, Pandas 3.0.1, NumPy 2.4.3, scikit-learn 1.8.0, Matplotlib 3.10.8, Seaborn 0.13.2. **Data source:** CoinMarketCap (https://coinmarketcap.com).